

MOHAMMED BALKHAIR

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🌐 Balkhair Mohammed

🐙 GitHub

🔗 Portfolio

Summary

CS Graduate & AI Specialist focused on end-to-end ML, DL, and GenAI (LLMs, GANs, Diffusion). Expertise in building and deploying scalable AI agents and architectures (Transformers, CNNs, GNNs) using Python, PyTorch, FastAPI, and Docker.

Education

Computer Science Bachelor's Degree

King Abdulaziz University, 4.83/5 GPA - First Honors

2021 – 2025

Professional Experience

AI Teaching Assistant, King Abdullah University of Science and Technology (KAUST)

- **Instructed 100+ students** in Advanced Computer Vision, including Object Detection and Generative AI.

01/2026 – 02/2026

Riyadh, Saudi Arabia

AI Research Trainee , Princess Margaret Cancer Center

- **Accelerated** drug discovery workflows by **13%** via prioritized high-throughput data pipelines.
- **Boosted throughput by 75%** by architecting **distributed inference** for GNN and Diffusion models.

06/2025 – 08/2025

Toronto, Canada

Data Scientist , Saudi Aramco (internship)

- **Architected** high-scale USPTO data pipelines and custom BERT classifiers (98% accuracy) for Aramco-specific IP mapping, reducing data acquisition cycles.
- **Integrated** these AI-driven insights into IP workflows, accelerating decision-making lead times by **20%** via automated reporting.

06/2024 – 08/2024

Dhahran, Saudi Arabia

Projects & Researches

DARIP : Daily AI-Generated Recaps in Podcast

- **GenAI Pipeline:** Integrated RSS scraping, RAG, and LLM-based dialogue generation with TTS synthesis for automated content creation.
- **Infrastructure & Deployment:** Leveraged **LangGraph** for multi-agent orchestration and **FastAPI/Docker** for scalable, cost-efficient deployment.

AI-LAB: Cloud-Based Machine Learning Model Hosting Platform

- **AI Model Hosting Platform:** Designed and deployed a scalable platform for hosting models via API using **FastAPI** and **Docker**.
- **Infrastructure Abstraction:** Streamlined AI experimentation by abstracting infrastructure management, allowing users to deploy and test models via a seamless API interface.

Development of MutDTA Models for Drug-Target Binding Affinity Prediction (Research)

- **Developed** multi-modal GNN-based DTA models for protein-ligand embeddings and 3D pose generation (GANs/ Diffusion).
- **Enhanced** binding-affinity accuracy for cancer proteins, directly supporting downstream lab experiments .

Certificates :

KAUST: AI Specialization (440 hrs).

DeepLearning.AI 📄 : Hyperparameter Tuning

SDAIA: AI professionals Bootcamp (210 hrs)

IBM: Back-End Dev Professional

Skills

Technical

- **Languages & Data:** Python, Java, SQL, NumPy, Pandas, Matplotlib.
- **AI Specialization:** ML, DL, CV, NLP, LLMs, GANs, Diffusion, GNNs, Transformers.
- **Frameworks:** PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face.

Soft Skills

English proficiency (IELTS : 6.5) , Problem Solving , Critical thinking, Quick learner , Time management , Teamwork , Self-Motivation

Development

FastAPI, Flask, LangGraph, Docker, DigitalOcean, Git